بِسۡمِ ٱللَّهِ ٱلرَّحۡمَٰنِ ٱلرَّحِيمِ

(Start the day with the name of Allah)

Welcome To Virtual University

Time Left sec(s)

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 7 of 10 (Start time: 10:03:27 PM, 06 June 2021)

Total Marks: 1

If f(x) = x cos(x), then which of the following is true about it?

Select the correct option

- (selected) Its derivative with respect to x is
-xsin(x) + cos(x).
- Its derivative with respect to x is
-xsin(x) + sin(x).
- None of these.
- Its derivative with respect to x is
xsin(x) + cos(x).

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 8 of 10 (Start time: 10:03:47 PM, 06 June 2021)

Total Marks: 1

If y=f(x) then instantaneous rate of change of y with respect to x at the point x0 is given by slope of tangent line at (x0,f(x0))

Select the correct option

- False
- True

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 9 of 10 (Start time: 10:04:04 PM, 06 June 2021)

Total Marks: 1

x^0 where $x \neq 0$ is equal to ………

Select the correct option

Reload Math Equations

- [x] 1
- [ ] None of these
- [ ] 0
- [ ] ∞

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 10 of 10 (Start time: 10:04:25 PM, 06 June 2021)

Total Marks: 1

If y = x^2, then which of the following is true about it over the interval [3, 4]
NOTE:- where x^n denotes the nth power of x.

Select the correct option

- (selected) Its average rate of change is 7.
- Its average rate of change is 9.
- None of these.
- Its average rate of change is 8.

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 1 of 10 (Start time: 08:31:09 AM, 06 June 2021)

Total Marks: 1

$The\ derivative\ of\ x^3\ is\ equal\ to..........?$

Select the correct option

Reload Math Equations

- ○ $3x^3$
- ● $3x^2$
- ○ x
- ○ 1

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 2 of 10 (Start time: 08:31:35 AM, 06 June 2021)

Total Marks: 1

Derivative of absolute value function:
f(x) = Abs(x) = |x| at x = -2 is

Select the correct

- -1
- does not exist
- zero
- -2

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 3 of 10 (Start time: 08:32:06 AM, 06 June 2021)

Total Marks: 1

f(x) = Cosx is differentiable in the interval ______.

Select the correct option

- (-inf, 0]
- (-inf,+inf)
- [0,+inf)
- All are true

Question # 4 of 10 (Start time: 08:32:27 AM, 06 June 2021) Total Marks: 1

What is the derivative $\tan(x^2 + x^3)$?

Select the correct option

Reload Math Equations

- ○ $(2x^3 + 3x)\sec^2(x^2 + x^3)$
- ○ $(2x^3 + 3x)\cos ec^2(x^2 + x^3)$
- ● $(3x^2 + 2x)\sec^2(x^2 + x^3)$
- ○ None of these

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 5 of 10 (Start time: 08:33:08 AM, 06 June 2021)

Total Marks: 1

If y = x^2, then which of the following is true about it over the interval [3, 4]
NOTE:- where x^n denotes the nth power of x.

Select the correct option

- Its average rate of change is 7.

- None of these.
- Its average rate of change is 8.
- Its average rate of change is 9.

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 6 of 10 (Start time: 08:33:33 AM, 06 June 2021)

Total Marks: 1

$$At\ x = \pi,\ the\ derivative\ of\ f(x) = a\cos x - b\sin x\ is$$
$$(note : \pi = 180 degrees)$$

Select the correct option

Reload Math Equations

○	-b
○	a
●	b
○	-a

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 7 of 10 (Start time: 08:33:59 AM, 06 June 2021)

Total Marks: 1

$$If\ f(x) = (x^2 + \sin(x) + \cos(x))^{10},\ then\ the\ derivative\ of\ f(x)\ is\ \_\_\_\_.$$

Select the correct option

Reload Math Equations

- $10(x^2 + \sin(x) + \cos(x))^9$
- *None of these.*
- $10(x^2 + \sin(x) + \cos(x))^9(2x + \cos(x) - \sin(x))$
- $10(x^2 + \sin(x) + \cos(x))^9(2x + \cos(x) + \sin(x)).$

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 8 of 10 (Start time: 08:34:32 AM, 06 June 2021)

Total Marks: 1

Which of the following rule will be used to find the derivative of $x\cot x$?

Select the correct option

Reload Math Equations

- sum
- power
- product
- difference

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 9 of 10 (Start time: 08:35:02 AM, 06 June 2021)

Total Marks: 1

If (0,0) and (2,2) are any two pints on a curve then the average speed of a particle passing through this curve from (0,0) to (2,2) is --------

Select the correct option

- -1
- 2
- -2
- 1

MTH101:Quiz 1

Question # 10 of 10 (Start time: 08:35:24 AM, 06 June 2021)

Simplifed expression for f(x)=tanx/secx is

Select the correct option

Sinx

Cosx

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 1 of 10 (Start time: 08:20:35 AM, 06 June 2021)

Total Marks: 1

For the curve: f(t) = -2t+3, the instantaneous rate of change of 'f(t)' at 't = a' is ______.

Select the correct option

- -1
- -2
- 1
- 2

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 2 of 10 (Start time: 08:21:05 AM, 06 June 2021)

Total Marks: 1

Slope of secant line joining points (1,1) and (3,4) is____.

Select the correct option

- 1.5
- 1
- 1.3
- None of these

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 3 of 10 (Start time: 08:22:27 AM, 06 June 2021)

Total Marks: 1

$$If\ f(x) = secx\ and\ g(x) = csc(x),\ then\ the\ derivative\ of\ f(x) \times g(x)\ is$$

Select the correct option

Reload Math Equations

- $-\sec x \csc x \cot x - \csc x \sec x \tan x$
- None of these.
- $-\sec x \csc x \cot x + \csc x \sec x \tan x$
- $\sec x \csc x \cot x + \csc x \sec x \tan x$

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 4 of 10 (Start time: 08:23:18 AM, 06 June 2021)

Total Marks: 1

Derivative of a straight line parallel to x-axis is------

Select the correct option

- -1
- Infinity
- +1
- Zero

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 5 of 10 (Start time: 08:23:57 AM, 06 June 2021)

Total Marks: 1

$(f.g)' = f'.g - f.g'$

Select the correct option

True

False

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 6 of 10 (Start time: 08:24:26 AM, 06 June 2021)

Total Marks: 1

The graph of function y=2 is a.....line with slope 0,is

Select the correct option

- Horizontal

- Vertical

Saving

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 7 of 10 (Start time: 08:25:16 AM, 06 June 2021)

Total Marks: 1

$$\text{If } u(t) = 4\cos t\,,\ then\ u'(\frac{\pi}{2}) = \ldots$$

Select the correct option

Reload Math Equations

- 2
- 9
- -4
- 4

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 8 of 10 (Start time: 08:26:00 AM, 06 June 2021)

Total Marks: 1

$$If\ f(x) = x^5\ and\ g(x) = x^2,\ then\ the\ derivative\ of\ \frac{f(x)}{g(x)}\ is$$

Select the correct option

Reload Math Equations

- ○ 3x
- ● $3x^2$
- ○ None of these
- ○ $\frac{3}{x^2}$

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 9 of 10 (Start time: 08:26:26 AM, 06 June 2021)

Total Marks: 1

What is the derivative of $\cot(x^6)$?

Select the correct option

Reload Math Equations

○	$\cos ec(x^6)\cot(x^6)$
○	None of these
●	$-6x^5\cos ec^2(x^6)$
○	$-\cos ec(x^6)$

MTH101:Quiz 1

Question # 10 of 10 (Start time: 08:26:57 AM, 06 June 2021)

$If\ f(x)=x^5\ and\ g(x)=x^2,\ then\ thederivative\ of\ f(x)\times g(x)\ is$

Select the correct option

○	$6x^2$
○	$3x^2$
●	None of these
○	$7x^2$

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 1 of 10 (Start time: 08:10:14 AM, 06 June 2021)

Total Marks: 1

What is the derivative of $\cos(2x)$ at $x = \pi/4$?

Select the correct option

Reload Math Equations

- 0
- None of these
- -2
- 2

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 2 of 10 (Start time: 08:11:02 AM, 06 June 2021)

Total Marks: 1

d(-2x+3cosx)/dx=..

Select the correct option

- 2-3sinx
- -2-3sinx
- -2+3sinx
- 2+3sinx

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 3 of 10 (Start time: 08:11:43 AM, 06 June 2021)

Total Marks: 1

If y = x^2, then which of the following is true about it over the interval [3, 4]
NOTE:- where x^n denotes the nth power of x.

Select the correct option

- None of these.
- Its average rate of change is 9.
- Its average rate of change is 7.
- Its average rate of change is 8.

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 4 of 10 (Start time: 08:12:14 AM, 06 June 2021)

Total Marks: 1

If in the graph of f(x) , there comes a corner then the function is -------- at that corner.

Select the correct option

- ○ Differentiable
- ● Non differentiable

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 5 of 10 (Start time: 08:12:46 AM, 06 June 2021)

Total Marks: 1

$$If\ y = f(\sin t)\ and\ y = t\ then\ \left(\frac{df}{dy}\right) = \ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

Select the correct option

Reload Math Equations

- $\sec y$
- $-\sec y$
- $\cos y$
- $-\cos y$

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 6 of 10 (Start time: 08:13:17 AM, 06 June 2021)

Total Marks: 1

Which of the followings is the derivative of f(x) = 2log x?

Select the correct option

Reload Math Equations

○	$f'(x) = \frac{2}{x^2}$
○	$f'(x) = \frac{1}{xln2}$
○	$f'(x) = \frac{1}{\sqrt{x}}$
●	$f'(x) = \frac{2}{x}$

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 7 of 10 (Start time: 08:14:01 AM, 06 June 2021)

Total Marks: 1

A bus travels 175 miles in 2.5 hours. Its average velocity will be.......

Select the correct option

- 80m/h
- 90m/h
- 70m/h
- 60m/h

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 8 of 10 (Start time: 08:14:38 AM, 06 June 2021)

Total Marks: 1

$f\ f(x) = \tan x^2$ *and* $g(x) = \sin x^{10}$, *then the derivative of* $f(x) + g(x)$ *is ....*

Select the correct option

Reload Math Equations

- ○ $\sec^2(x^2) + \cos x^{10}$
- ○ $2x\sec(x^2) + 10x^9\cos x^{10}$
- ● $2x\sec^2(x^2) + 10x^9\cos x^{10}$
- ○ None of these

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 9 of 10 (Start time: 08:15:36 AM, 06 June 2021)

Total Marks: 1

$$Derivative\ of\ \sin(5x+3)\ is\ \ldots\ldots\ldots\ldots\ldots\ldots$$

Select the correct option

Reload Math Equations

○	$3\cos(5x+3)$
○	$3\cos 5x$
○	$-5\sin(5x+3)$
	$5cox(5x+3)$

MTH101:Quiz 1

Question # 10 of 10 (Start time: 08:16:13 AM, 06 June 2021)

Secx is equal to the reciprocal of

Select the correct option

- [ ] Cosecx
- [ ] Tanx
- [x] Cosx
- [ ] Sinx

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 1 of 10 (Start time: 09:42:01 AM, 06 June 2021)

Total Marks: 1

If the function is differentiable at a point then it is also _____ at that point.

Select the correct option

- [x] Continuous
- [ ] Not defined

MTH101:Quiz 1 Quiz Start Time: 09:42 AM, 06 June 2021

Question # 2 of 10 (Start time: 09:42:41 AM, 06 June 2021) Total Marks: 1

The derivative of function:
f(x) = -2(x-1) + {5/(x-1)} does not exist at
x = --------

Select the correct option

- -5
- -1
- 1
- 0

sec(s)

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 3 of 10 (Start time: 09:43:18 AM, 06 June 2021)

Total Marks: 1

If a truck travels 100 miles over a straight road in a 5-hour period, then the average velocity of the truck will be

Select the correct option

- ○ None of these
- ○ 12 miles/hour
- ○ 5 miles/hour
- ● 20 miles/hour

MTH101:Quiz 1

Quiz Start Time: 09:42 AM.

Question # 4 of 10 (Start time: 09:43:54 AM, 06 June 2021)

The power rule for derivative says to _____ '1' from the exponent.

Select the correct option

- Divide
- Subtract
- Multiply
- Add

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 5 of 10 (Start time: 09:44:19 AM, 06 June 2021)

Total Marks: 1

If f (x) = 1 / x, then which of the following is true about it.

Select the correct option

- None of these.
- Its derivative with respect to x is (- x^2).
- Its derivative with respect to x is (-1 /x^2).
- Its derivative with respect to x is (1 / x^2).

Saving...

sec(s)

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 6 of 10 (Start time: 09:44:38 AM, 06 June 2021)

Total Marks: 1

$$If\ f(x) = \sqrt{x}\ and\ g(x) = 4\sqrt{x}, then\ the\ derivative\ of\ f(x) \times g(x)\ is$$

Select the correct option

Reload Math Equations

●	4
○	None of these
○	x^2
○	-4

Saving...

sec(s)

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 7 of 10 (Start time: 09:45:02 AM, 06 June 2021)

Total Marks: 1

Velocity is the rate of change of position w.r.t ....

Select the correct option

- time
- displacement
- acceleration
- force

Saving...

sec(s)

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 8 of 10 (Start time: 09:45:29 AM, 06 June 2021)

Total Marks: 1

(f.g)' = f '.g -f.g'

Select the correct option

- True
- False

sec(s)

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 9 of 10 (Start time: 09:45:47 AM, 06 June 2021)

Total Marks: 1

Slope of secant line joining points (1,1) and (3,4) is____.

Select the correct option

- 1.5
- 1
- None of these
- 1.3

sec(s)

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 1 of 10 (Start time: 05:10:20 PM, 10 June 2021)

Total Marks: 1

Slope of the graph of the function: $f(x) = -4x+99$ at the point (99,100) is ----

Select the correct option

- ○ 99
- ○ -1
- ○ 100
- ● -4

Question # 2 of 10 (Start time: 05:10:38 PM, 10 June 2021) Total Marks: 1

What is the derivative of cos (45)?

Select the correct option

- -sin(45)
- Sec(45)
- sin(45)
- None of these

sec(s)

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 3 of 10 (Start time: 05:12:11 PM, 10 June 2021)

Total Marks: 1

In the interval [0,+2pi], the minimum value of derivative of 'f(x) = Sinx' exists at x = ----- (note: pi=180 degrees)

Select the correct option

- ○ pi/2
- ○ 0
- ○ pi/4
- ● pi

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 4 of 10 (Start time: 05:12:36 PM, 10 June 2021)

Total Marks: 1

If f(t) =8t-(5/t) then f'(t)=______

Select the correct option

- 8t+(5/t^2)
- 8+(5/t)
- 8+(5/t^2)
- 8-(5/t)

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 5 of 10 (Start time: 05:13:02 PM, 10 June 2021)

Total Marks: 1

$g(x) = |x+3|$ *is the composition of two functions , one is* $(x+3)$ *and the other is ....*

Select the correct option

Reload Math Equations

- ○ 0
- ○ x
- ● $|x|$
- ○ 3

بسم اللہ الرحمن الرحیم

VU KMS

اسلام علیکم:

امید کرتا ہوں سب خیریت سے ہوں گئے۔ آنے والے تمام نیو طالب علم کو میں ورچوئل یونیورسٹی آف پاکستان میں خوش آمدید کرتا ہوں۔

اگر کسی سٹوڈنٹ کو کسی بھی قسم کی ہیلپ چاہے ہو تو ہم سے رابط کر سکتا ہے جیسا کہ: پاسٹ پیپرز، کورس سلیکشن، پروجیکٹ سلیکشن، ڈیٹ شیٹ، فیس پے اور چینج سٹڈی پروگرام وغیرہ۔

Paid VU LMS

اگر کوئی سٹوڈنٹ کسی مجبوری کی وجہ سے یا جاب ہولڈر ہونے کی وجہ سے اپنا ایل ایم ایس خود ہینڈل نہیں کر سکتا تو وہ ہم سے پیڈ سروس حاصل کر سکتا ہے کوشش کرے کے آپ ہر چیز خود اٹیمپٹ کریں اگر نہیں کر سکتے تو سروس حاصل کریں۔

انشاءاللہ آپ ہماری سروس سے مطمئن ہوں گے۔

Email:
waqasahmeduog@gmail.com

مزید اگر کسی رہنمائی کی ضرورت ہو تو وہ رابطہ کر سکتا ہے۔

Whatsapp: 03066295623 , 03404643855

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 6 of 10 (Start time: 05:13:22 PM, 10 June 2021)

Total Marks: 1

The power rule for derivative says to _____ '1' from the exponent.

Select the correct option

- ○ Divide
- ○ Add
- ● Subtract
- ○ Multiply

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 7 of 10 (Start time: 05:13:44 PM, 10 June 2021)

Total Marks: 1

the derivative of 1/(x+1) is

Select the correct option

- ○ -1/(x-1)
- ○ 1/x
- ○ 1/(x+1)
- ● -1/(x+1).(x+1)

Question # 8 of 10 (Start time: 05:14:03 PM, 10 June 2021)

Total Marks: 1

What is the derivative of

$$39x^{\frac{14}{13}}$$

Select the correct option

Reload Math Equations

- ○ $24x^{\frac{-1}{13}}$
- ● $42x^{\frac{1}{13}}$
- ○ None of these
- ○ $42x$

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 9 of 10 (Start time: 05:14:29 PM, 10 June 2021)

Total Marks: 1

Let a car moves from lower mall with the average velocity of 100 km/h and after covering the distance 50 km , the car reach at the airport Lahore. The time elapsed is

Select the correct option

- [x] 1/2
- [ ] 2
- [ ] -1/2
- [ ] None of these

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 10 of 10 (Start time: 05:14:52 PM, 10 June 2021)

Total Marks: 1

$$If\ f(x)=\sqrt{x}\ and\ g(x)=4\sqrt{x},\ then\ the\ derivative\ of\ \frac{f(x)}{g(x)}\ is$$

Select the correct option

Reload Math Equations

- 2
- 4
- None of these
- x + 2

MTH101:Quiz 1 Quiz Start Time: 10:01 PM, 06 June 2021

Question # 1 of 10 (Start time: 10:01:11 PM, 06 June 2021) Total Marks: 1

$$If\ f(x) = 5\sin(x+1)\ then\ its\ derivative\ by\ chain\ rule\ at\ x = -1\ is \ldots\ldots\ldots\ldots$$

Select the correct option

Reload Math Equations

○	6
○	-1
○	1
●	5

Question # 1 of 10 (Start time: 05:48:03 PM, 08 June 2021) Total Marks: 1

$If\ f(x) = \cot^5(x^{20}),\ then\ which\ of\ the\ following\ is\ true\ about\ it$

Select the correct option

Reload Math Equations

$Its\ derivative\ with\ respect\ to\ x\ is\ -100x^{19}\cot^4(x^{20})\cos ec^2(x^{20})$

None of these

$Its\ derivative\ with\ respect\ to\ x\ is\ x^{19}\cot^4(x^{20})\cos ec^2(x^{20})$

$Its\ derivative\ with\ respect\ to\ x\ is\ 100x^{19}\cot^4(x^{20})\cos ec^2(x^{20})$

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 2 of 10 (Start time: 05:48:28 PM, 08 June 2021)

Total Marks: 1

If (0,0) and (2,2) are any two points on a curve then the slope of a line parallel to the secant segment through these points is ________.

Select the correct option

- ○ 2
- ○ -1
- ● 1
- ○ -2

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 3 of 10 (Start time: 05:48:46 PM, 08 June 2021) **Total Marks: 1**

If the slope of a function is zero, then which of the following is true about it.

Select the correct option

- ○ None of these.
- ● The function should be a constant.
- ○ The function should be a linear.
- ○ The function should be a quadratic.

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 4 of 10 (Start time: 05:49:04 PM, 08 June 2021)

Total Marks: 1

If $f(x) = \sin^4(x^2)$ *, then the derivative of* $f(x)$ *is* ______.

Select the correct option

Reload Math Equations

- None of these.
- $8x\sin^3(x^2)\cos(x).$
- $x\sin^3(x^2)\cos(x^2).$
- $8x\sin^3(x^2).$

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 5 of 10 (Start time: 05:49:25 PM, 08 June 2021)

Total Marks: 1

When the slope of the tangent line approaches to ±infinity , then ...... occurs

Select the correct option

- [ ] circle
- [ ] horizontal tangent
- [x] vertical tangents
- [ ] corners

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 6 of 10 (Start time: 05:49:44 PM, 08 June 2021)

Total Marks: 1

$$If\ x = f(-3y)\ and\ y = 2x\ then\ \frac{df}{dx} = \ldots\ldots\ldots\ldots\ldots$$

Select the correct option

Reload Math Equations

- -1/6
- -6
- -6
- 6

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 7 of 10 (Start time: 05:50:03 PM, 08 June 2021)

Total Marks: 1

On the straight line, the tangent at any point coincide with line -------

Select the correct option

- some where
- every where
- none of these
- no where

Question # 8 of 10 (Start time: 05:50:19 PM, 08 June 2021) Total Marks: 1

The derivative of the sum is equal to the ....of the derivatives

Select the correct option

- ○ Product
- ○ Quotient
- ○ Difference
- ● Sum

Question # 9 of 10 (Start time: 05:50:39 PM, 08 June 2021)

Total Marks: 1

$$If\ f(x) = \cot(2x),\ then\ the\ derivative\ of\ f(x) is$$

Select the correct option

Reload Math Equations

- None of these.
- $-2csc^2(2x)$
- $2csc^2(2x)$
- $-2\csc(2x)$

Question # 10 of 10 (Start time: 05:50:57 PM, 08 June 2021) **Total Marks: 1**

Derivative of a straight line parallel to y-axis is----

Select the correct option

- ○ Zero
- ○ -1
- ○ +1
- ● Infinity

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 2 of 10 (Start time: 10:01:37 PM, 06 June 2021)

Total Marks: 1

At a corner a tangent line does not exist because the slopes of the secant lines do not have a (two sided) limit.

Select the correct option

- ○ False
- ● True

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 3 of 10 (Start time: 10:01:54 PM, 06 June 2021)

Total Marks: 1

$$\frac{d}{dx}(-x\sin x) =$$

Select the correct option

Reload Math Equations

- -xCosx-Sinx
- -Sinx-Cosx
- Sinx-xCosx
- -xCosx+Sinx

sec(s)

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 1 of 10 (Start time: 06:20:37 PM, 06 June 2021)

Total Marks: 1

$$The\ derivative\ of\ (x^2) + 2sinx\ w.r.t\ x\ is$$

Select the correct option

Reload Math Equations

- [x] 2x+2cosx

- [ ] 2x-2cosx
- [ ] -2x-2cosx
- [ ] -2x+2sinx

sec(s)

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 2 of 10 (Start time: 06:21:06 PM, 06 June 2021)

Total Marks: 1

Value of the derivative of 'f(x) = Tanx' at
x = pi, is -------
(note: pi=180 degrees)

Select the correct option

- 0
- 1/2
- -1
- 1

Click to Save Answer & Move to Next Question

sec(s)

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 3 of 10 (Start time: 06:21:30 PM, 06 June 2021)

Total Marks: 1

Derivative of a constant is zero.

Select the correct option

- [x] True
- [ ] False

Click to Save Answer & Move to Next Question

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 4 of 10 (Start time: 06:21:51 PM, 06 June 2021)

Total Marks: 1

If f (x) = x^3 + 1, then which of the following is NOT true about it.
NOTE:- where x^n denotes the nth power of x .

Select the correct option

- ○ The slope of tangent line at x = -2 is 12.
- ○ The slope of tangent line at x = 2 is 12.
- ● The slope of tangent line at x = 1 is 3.
- ○ None of these. .

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 5 of 10 (Start time: 06:22:13 PM, 06 June 2021)

Total Marks: 1

Average velocity is obtained by divding the ....

Select the correct option

- Total distance by time elapsed
- Total distance by velocity elapsed

sec(s)

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 6 of 10 (Start time: 06:22:33 PM, 06 June 2021)

Total Marks: 1

If f(x) = 8x^2 -2x +5 , then f ' (0) will be......

Select the correct option

- ○ 4
- ● -2
- ○ -1
- ○ -3

Saving...

sec(s)

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 7 of 10 (Start time: 06:22:54 PM, 06 June 2021)

Total Marks: 1

Derivative of a straight line parallel to y-axis is----

Select the correct option

- -1
- Zero
- Infinity
- +1

sec(s)

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 8 of 10 (Start time: 06:23:13 PM, 06 June 2021)

Total Marks:

If a car travels 75 miles over a straight line in a 3-hour period, then we say that the average velocity of the car is.....miles per hour

Select the correct option

- 3.75
- 75
- 75/3
- 3/75

Saving.

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 9 of 10 (Start time: 06:23:34 PM, 06 June 2021)

Total Marks: 1

Derivative of function: f(x) = Sin(x), exits-------

Select the correct option

- everywhere -
- only at origin
- nowhere
- somewhere

Click to Save Answer & Move to Next Question

MTH101:Quiz 1

Question # 10 of 10 (Start time: 06:23:56 PM, 06 June 2021)

The equation of tangent line at the point P(2,3) to the curve y = 3 x^2 having slope 12 is______.

Select the correct option

- ○ y-12 x - 21 = 0
- ● y-12 x + 21 = 0
- ○ y+12 x + 21 = 0
- ○ None of these.

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 4 of 10 (Start time: 10:02:16 PM, 06 June 2021)

Total Marks: 1

Let y = x^2+1
Then the instantaneous rate of change of y with respect to x at the point -2 is

Select the correct option

- ○ 0
- ● -4
- ○ 3
- ○ -1

sec(s)

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 1 of 10 (Start time: 05:06:25 PM, 07 June 2021)

Total Marks: 1

$$If\ f(x) = \cos^4(x^2),\ then\ the\ derivative\ of\ f(x)\ is\ \_\_\_\_\_.$$

Select the correct option

Reload Math Equations

- (selected) $-8x\cos^3(x^2)\sin(x^2)$
- $8x\cos^3(x^2)\sin(x^2)$.
- *None of these.*
- $x\cos^3(x^2)\sin(x^2)$

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 2 of 10 (Start time: 05:07:01 PM, 07 June 2021)

Total Marks: 1

On the straight line, the tangent at any point coincide with line -------

Select the correct option

- every where
- no where
- some where
- none of these

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 3 of 10 (Start time: 05:07:47 PM, 07 June 2021)

Total Marks: 1

If y = x^2, then which of the following is true about it over the interval [3, 4]
NOTE:- where x^n denotes the nth power of x.

Select the correct option

- [x] Its average rate of change is 7.
- [ ] Its average rate of change is 8.
- [ ] None of these.
- [ ] Its average rate of change is 9.

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 4 of 10 (Start time: 05:08:10 PM, 07 June 2021)

Total Marks: 1

If f (x) = 1 / x, then which of the following is true about it.

Select the correct option

- ( ) Its derivative with respect to x is (1 / x^2).
- (•) Its derivative with respect to x is (-1 /x^2).
- ( ) Its derivative with respect to x is (- x^2).
- ( ) None of these.

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 5 of 10 (Start time: 05:08:30 PM, 07 June 2021)

Total Marks: 1

$If\ f(x) = secx\ and\ g(x) = csc(x),\ then\ the\ derivative\ of\ f(x) \times g(x)\ is$

Select the correct option

Reload Math Equations

- ○ $\sec x \csc x \cot x + \csc x \sec x \tan x$
- ○ None of these.
- ● $-\sec x \csc x \cot x + \csc x \sec x \tan x$
- ○ $-\sec x \csc x \cot x - \csc x \sec x \tan x$

Saving...

sec(s)

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 6 of 10 (Start time: 05:08:59 PM, 07 June 2021)

Total Marks: 1

The derivative of y=Cosecx at x=0 is

Select the correct option

- 1
- -1
- 0
- Infinity

VU KMS

Whatsapp: 03066295623

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 7 of 10 (Start time: 05:09:17 PM, 07 June 2021)

Total Marks: 1

$$Derivative\ of\ (y+1)\ w.r.t\ 'x'\ is.....................$$

Select the correct option

Reload Math Equations

●	$\frac{dy}{dx}$
○	Does not exist
○	1
○	$\frac{dx}{dy}$

Click to Save Answer & Move to Next Question

sec(s)

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 8 of 10 (Start time: 05:09:43 PM, 07 June 2021)

Total Marks: 1

Derivative of x is 1

Select the correct option

- [x] True
- [ ] False

Question # 9 of 10 (Start time: 05:10:56 PM, 07 June 2021) Total Marks: 1

$$If\ f(x) = \tan^2(x^5)\ ,\ then\ the\ derivative\ of\ f(x)\ is\ \_\_\_\_\_.$$

Select the correct option

Reload Math Equations

- None of these.
- $10x^4\sec^2(x^5)\tan(x^5).$
- $-10x^4\sec^2(x^5)\tan(x^5).$
- $x^4\sec^2(x^5)\tan(x^5).$

MTH101:Quiz 1

Question # 10 of 10 (Start time: 05:11:25 PM, 07 June 2021)

pi radians corresponds to..........degrees.

Select the correct option

- 75
- 360
- 180
- 90

Click t

Question # 5 of 10 (Start time: 10:02:40 PM, 06 June 2021)

Total Marks: 1

$$What\ is\ the\ derivative\ of\ (3x^2+1)^{52}?$$

Select the correct option

Reload Math Equations

- ○ $(6x)^{52}$
- ● $312x(3x^2+1)^{51}$
- ○ $52(3x^2+1)^{51}$
- ○ *None of these*

sec(s)

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 6 of 10 (Start time: 10:03:08 PM, 06 June 2021)

Total Marks: 1

If a function f is continuous at a point then it ________________ be differentiable at that point.

Select the correct option

- ○ Is necessarily
- ● May

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 1 of 10 (Start time: 09:42:01 AM, 06 June 2021)

Total Marks: 1

If the function is differentiable at a point then it is also _____ at that point.

Select the correct option

- Continuous

- Not defined

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 2 of 10 (Start time: 09:42:41 AM, 06 June 2021)

Total Marks: 1

The derivative of function:
f(x) = -2(x-1) + {5/(x-1)} does not exist at
x = ---------

Select the correct option

- -5
- -1
- 1
- 0

sec(s)

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 3 of 10 (Start time: 09:43:18 AM, 06 June 2021)

Total Marks: 1

If a truck travels 100 miles over a straight road in a 5-hour period, then the average velocity of the truck will be

Select the correct option

- None of these
- 12 miles/hour
- 5 miles/hour
- 20 miles/hour

MTH101:Quiz 1

Quiz Start Time: 09:42 AM.

Question # 4 of 10 (Start time: 09:43:54 AM, 06 June 2021)

The power rule for derivative says to _____ '1' from the exponent.

Select the correct option

○	Divide
●	Subtract
○	Multiply
○	Add

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 5 of 10 (Start time: 09:44:19 AM, 06 June 2021)

Total Marks: 1

If f (x) = 1 / x, then which of the following is true about it.

Select the correct option

- None of these.
- Its derivative with respect to x is (- x^2).
- Its derivative with respect to x is (-1 /x^2).
- Its derivative with respect to x is (1 / x^2).

Saving...

sec(s)

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 6 of 10 (Start time: 09:44:38 AM, 06 June 2021)

Total Marks: 1

$$If\ f(x)=\sqrt{x}\ and\ g(x)=4\sqrt{x},\ then\ the\ derivative\ of\ f(x)\times g(x)\ is$$

Select the correct option

Reload Math Equations

●	4
○	None of these
○	x^2
○	-4

Saving...

sec(s)

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 7 of 10 (Start time: 09:45:02 AM, 06 June 2021)

Total Marks: 1

Velocity is the rate of change of position w.r.t ....

Select the correct option

- [x] time
- [ ] displacement
- [ ] acceleration
- [ ] force

Saving...

sec(s)

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 8 of 10 (Start time: 09:45:29 AM, 06 June 2021)

Total Marks: 1

(f.g)' = f '.g -f.g'

Select the correct option

True

False

VU KMS

Whatsapp: 03066295623

sec(s)

MTH101:Quiz 1

Quiz Start Time: 09:42 AM, 06 June 2021

Question # 9 of 10 (Start time: 09:45:47 AM, 06 June 2021)

Total Marks: 1

Slope of secant line joining points (1,1) and (3,4) is____.

Select the correct option

- [x] 1.5

- [ ] 1
- [ ] None of these
- [ ] 1.3

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 1 of 10 (Start time: 05:10:20 PM, 10 June 2021)

Total Marks: 1

Slope of the graph of the function: $f(x) = -4x+99$ at the point (99,100) is ----

Select the correct option

- ○ 99
- ○ -1
- ○ 100
- ● -4

Question # 2 of 10 (Start time: 05:10:38 PM, 10 June 2021) Total Marks: 1

What is the derivative of cos (45)?

Select the correct option

- -sin(45)
- Sec(45)
- sin(45)
- None of these

sec(s)

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 3 of 10 (Start time: 05:12:11 PM, 10 June 2021)

Total Marks: 1

In the interval [0,+2pi], the minimum value of derivative of 'f(x) = Sinx' exists at x = ----- (note: pi=180 degrees)

Select the correct option

- ○ pi/2
- ○ 0
- ○ pi/4
- ● pi

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 4 of 10 (Start time: 05:12:36 PM, 10 June 2021)

Total Marks: 1

If f(t) =8t-(5/t) then f(t)=______

Select the correct option

- 8t+(5/t^2)
- 8+(5/t)
- 8+(5/t^2)
- 8-(5/t)

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 5 of 10 (Start time: 05:13:02 PM, 10 June 2021)

Total Marks: 1

$g(x) = |x+3|$ *is the composition of two functions , one is* $(x+3)$ *and the other is ....*

Select the correct option

Reload Math Equations

- ○ 0
- ○ x
- ● $|x|$
- ○ 3

Question # 6 of 10 (Start time: 05:13:22 PM, 10 June 2021)

Total Marks: 1

The power rule for derivative says to _____ '1' from the exponent.

Select the correct option

- Divide
- Add
- Subtract
- Multiply

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 7 of 10 (Start time: 05:13:44 PM, 10 June 2021)

Total Marks: 1

the derivative of 1/(x+1) is

Select the correct option

- ○ -1/(x-1)
- ○ 1/x
- ○ 1/(x+1)
- ● -1/(x+1).(x+1)

MTH101:Quiz 1

Quiz Start Time: 05:10 PM, 10 June 2021

Question # 8 of 10 (Start time: 05:14:03 PM, 10 June 2021)

Total Marks: 1

What is the derivative of

$$39x^{\frac{14}{13}}$$

Select the correct option

Reload Math Equations

- ○ $24x^{\frac{1}{13}}$
- ● $42x^{\frac{1}{13}}$
- ○ None of these
- ○ $42x$

MTH101:Quiz 1 Quiz Start Time: 05:10 PM, 10 June 2021

Question # 9 of 10 (Start time: 05:14:29 PM, 10 June 2021) Total Marks: 1

Let a car moves from lower mall with the average velocity of 100 km/h and after covering the distance 50 km , the car reach at the airport Lahore. The time elapsed is

Select the correct option

- 1/2
- 2
- -1/2
- None of these

MTH101:Quiz 1 Quiz Start Time: 05:10 PM, 10 June 2021

Question # 10 of 10 (Start time: 05:14:52 PM, 10 June 2021) Total Marks: 1

$$If\ f(x)=\sqrt{x}\ and\ g(x)=4\sqrt{x},\ then\ the\ derivative\ of\ \frac{f(x)}{g(x)}\ is$$

Select the correct option

Reload Math Equations

- ○ 2
- ○ 4
- ● None of these
- ○ x + 2

MTH101:Quiz 1 Quiz Start Time: 10:01 PM, 06 June 2021

Question # 1 of 10 (Start time: 10:01:11 PM, 06 June 2021) Total Marks: 1

$$If\ f(x) = 5\sin(x+1)\ then\ its\ derivative\ by\ chain\ rule\ at\ x = -1\ is \ldots\ldots\ldots\ldots$$

Select the correct option

Reload Math Equations

- ○ 6
- ○ -1
- ○ 1
- ● 5

Question # 1 of 10 (Start time: 05:48:03 PM, 08 June 2021) Total Marks: 1

$If\ f(x) = \cot^5(x^{20}),\ then\ which\ of\ the\ following\ is\ true\ about\ it$

Select the correct option

Reload Math Equations

$Its\ derivative\ with\ respect\ to\ x\ is\ -100x^{19}\cot^4(x^{20})\cos ec^2(x^{20})$

None of these

$Its\ derivative\ with\ respect\ to\ x\ is\ x^{19}\cot^4(x^{20})\cos ec^2(x^{20})$

$Its\ derivative\ with\ respect\ to\ x\ is\ 100x^{19}\cot^4(x^{20})\cos ec^2(x^{20})$

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 2 of 10 (Start time: 05:48:28 PM, 08 June 2021)

Total Marks: 1

If (0,0) and (2,2) are any two points on a curve then the slope of a line parallel to the secant segment through these points is ________.

Select the correct option

- ○ 2
- ○ -1
- ● 1
- ○ -2

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 3 of 10 (Start time: 05:48:46 PM, 08 June 2021)

Total Marks: 1

If the slope of a function is zero, then which of the following is true about it.

Select the correct option

- ○ None of these.
- ● The function should be a constant.
- ○ The function should be a linear.
- ○ The function should be a quadratic.

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 4 of 10 (Start time: 05:49:04 PM, 08 June 2021)

Total Marks: 1

$$If\ f(x) = \sin^4(x^2)\ ,\ then\ the\ derivative\ of\ f(x)\ is\ \_\_\_\_\_.$$

Select the correct option

Reload Math Equations

- [x] None of these.
- [ ] $8x\sin^3(x^2)\cos(x)$.
- [ ] $x\sin^3(x^2)\cos(x^2)$.
- [ ] $8x\sin^3(x^2)$.

Question # 5 of 10 (Start time: 05:49:25 PM, 08 June 2021) Total Marks: 1

When the slope of the tangent line approaches to ±infinity , then ..... occurs

Select the correct option

- ○ circle
- ○ horizontal tangent
- ● vertical tangents
- ○ corners

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 6 of 10 (Start time: 05:49:44 PM, 08 June 2021)

Total Marks: 1

$$If\ x = f(-3y)\ and\ y = 2x\ then\ \frac{df}{dx} = \ldots\ldots\ldots\ldots\ldots\ldots$$

Select the correct option

Reload Math Equations

- -1/6
- -6
- -6
- 6

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 7 of 10 (Start time: 05:50:03 PM, 08 June 2021)

Total Marks: 1

On the straight line, the tangent at any point coincide with line --------

Select the correct option

- ○ some where
- ● every where
- ○ none of these
- ○ no where

Question # 8 of 10 (Start time: 05:50:19 PM, 08 June 2021) Total Marks: 1

The derivative of the sum is equal to the .... of the derivatives

Select the correct option

- ○ Product
- ○ Quotient
- ○ Difference
- ● Sum

MTH101:Quiz 1

Quiz Start Time: 05:48 PM, 08 June 2021

Question # 9 of 10 (Start time: 05:50:39 PM, 08 June 2021)

Total Marks:

$If\ f(x) = \cot(2x),\ then\ the\ derivative\ of\ f(x) is$

Select the correct option

Reload Math Equations

- None of these.
- $-2csc^2(2x)$
- $2csc^2(2x)$
- $-2\csc(2x)$

Question # 10 of 10 (Start time: 05:50:57 PM, 08 June 2021) **Total Marks: 1**

Derivative of a straight line parallel to y-axis is----

Select the correct option

- Zero
- -1
- +1
- Infinity

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 2 of 10 (Start time: 10:01:37 PM, 06 June 2021)

Total Marks: 1

At a corner a tangent line does not exist because the slopes of the secant lines do not have a (two sided) limit.

Select the correct option

- False
- True

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 3 of 10 (Start time: 10:01:54 PM, 06 June 2021)

Total Marks: 1

$$\frac{d}{dx}(-x\sin x) =$$

Select the correct option

Reload Math Equations

- [x] -xCosx-Sinx
- [ ] -Sinx-Cosx
- [ ] Sinx-xCosx
- [ ] -xCosx+Sinx

sec(s)

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 1 of 10 (Start time: 06:20:37 PM, 06 June 2021)

Total Marks: 1

$$The\ derivative\ of\ (x^2) + 2sinx\ w.r.t\ x\ is$$

Select the correct option

Reload Math Equations

- [x] 2x+2cosx
- [ ] 2x-2cosx
- [ ] -2x-2cosx
- [ ] -2x+2sinx

sec(s)

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 2 of 10 (Start time: 06:21:06 PM, 06 June 2021)

Total Marks: 1

Value of the derivative of 'f(x) = Tanx' at
x = pi, is -------
(note: pi=180 degrees)

Select the correct option

- ○ 0
- ○ 1/2
- ○ -1
- ● 1

Click to Save Answer & Move to Next Question

sec(s)

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 3 of 10 (Start time: 06:21:30 PM, 06 June 2021)

Total Marks: 1

Derivative of a constant is zero.

Select the correct option

- [x] True
- [ ] False

Click to Save Answer & Move to Next Question

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 4 of 10 (Start time: 06:21:51 PM, 06 June 2021)

Total Marks: 1

If f (x) = x^3 + 1, then which of the following is NOT true about it.
NOTE:- where x^n denotes the nth power of x .

Select the correct option

- ○ The slope of tangent line at x = -2 is 12.
- ○ The slope of tangent line at x = 2 is 12.
- ● The slope of tangent line at x = 1 is 3.
- ○ None of these. .

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 5 of 10 (Start time: 06:22:13 PM, 06 June 2021)

Total Marks: 1

Average velocity is obtained by divding the ....

Select the correct option

- Total distance by time elapsed

- Total distance by velocity elapsed

sec(s)

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 6 of 10 (Start time: 06:22:33 PM, 06 June 2021)

Total Marks: 1

If f(x) = 8x^2 -2x +5 , then f ' (0) will be.......

Select the correct option

- ○ 4
- ● -2
- ○ -1
- ○ -3

Saving...

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 7 of 10 (Start time: 06:22:54 PM, 06 June 2021)

Total Marks: 1

Derivative of a straight line parallel to y-axis is----

Select the correct option

- -1
- Zero
- Infinity
- +1

sec(s)

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 202

Question # 8 of 10 (Start time: 06:23:13 PM, 06 June 2021)

Total Marks:

If a car travels 75 miles over a straight line in a 3-hour period, then we say that the average velocity of the car is.....miles per hour

Select the correct option

- ○ 3,75
- ○ 75
- ● 75/3
- ○ 3/75

Saving.

MTH101:Quiz 1

Quiz Start Time: 06:20 PM, 06 June 2021

Question # 9 of 10 (Start time: 06:23:34 PM, 06 June 2021) **Total Marks: 1**

Derivative of function: $f(x) = \text{Sin}(x)$, exits-------

Select the correct option

- (●) everywhere -
- ( ) only at origin
- ( ) nowhere
- ( ) somewhere

Click to Save Answer & Move to Next Question

MTH101:Quiz 1

Question # 10 of 10 (Start time: 06:23:56 PM, 06 June 2021)

The equation of tangent line at the point P(2,3) to the curve y = 3 x^2 having slope 12 is______.

Select the correct option

- ○ y-12 x - 21 = 0
- ● y-12 x + 21 = 0
- ○ y+12 x + 21 = 0
- ○ None of these.

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 4 of 10 (Start time: 10:02:16 PM, 06 June 2021)

Total Marks: 1

Let y = x^2+1
Then the instantaneous rate of change of y with respect to x at the point -2 is

Select the correct option

- ○ 0
- ● -4
- ○ 3
- ○ -1

sec(s)

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 1 of 10 (Start time: 05:06:25 PM, 07 June 2021)

Total Marks: 1

$If\ f(x) = \cos^4(x^2),\ then\ the\ derivative\ of\ f(x)\ is$ ______.

Select the correct option

Reload Math Equations

- (selected) $-8x\cos^3(x^2)\sin(x^2)$
- $8x\cos^3(x^2)\sin(x^2)$.
- $None\ of\ these.$
- $x\cos^3(x^2)\sin(x^2)$

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 2 of 10 (Start time: 05:07:01 PM, 07 June 2021)

Total Marks: 1

On the straight line, the tangent at any point coincide with line -------

Select the correct option

- (selected) every where
- no where
- some where
- none of these

MTH101:Quiz 1 Quiz Start Time: 05:06 PM, 07 June 2021

Question # 3 of 10 (Start time: 05:07:47 PM, 07 June 2021) Total Marks: 1

If y = x^2, then which of the following is true about it over the interval [3, 4]
NOTE:- where x^n denotes the nth power of x.

Select the correct option

- (selected) Its average rate of change is 7.
- Its average rate of change is 8.
- None of these.
- Its average rate of change is 9.

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 4 of 10 (Start time: 05:08:10 PM, 07 June 2021)

Total Marks: 1

If f (x) = 1 / x, then which of the following is true about it.

Select the correct option

- ○ Its derivative with respect to x is (1 / x^2).
- ● Its derivative with respect to x is (-1 /x^2).
- ○ Its derivative with respect to x is (- x^2).
- ○ None of these.

MTH101:Quiz 1 Quiz Start Time: 05:06 PM, 07 June 2021

Question # 5 of 10 (Start time: 05:08:30 PM, 07 June 2021) Total Marks: 1

$$If\ f(x) = secx\ and\ g(x) = csc(x),\ then\ the\ derivative\ of\ f(x) \times g(x)\ is$$

Select the correct option

Reload Math Equations

- ( ) $\sec x \csc x \cot x + \csc x \sec x \tan x$
- ( ) None of these.
- (•) $-\sec x \csc x \cot x + \csc x \sec x \tan x$
- ( ) $-\sec x \csc x \cot x - \csc x \sec x \tan x$

Saving...

sec(s)

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 6 of 10 (Start time: 05:08:59 PM, 07 June 2021)

Total Marks: 1

The derivative of y=Cosecx at x=0 is

Select the correct option

- [ ] 1
- [ ] -1
- [ ] 0
- [x] Infinity

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 7 of 10 (Start time: 05:09:17 PM, 07 June 2021)

Total Marks: 1

$Derivative\ of\ (y+1)\ w.r.t\ 'x'\ is...................$

Select the correct option

Reload Math Equations

●	$\frac{dy}{dx}$
○	Does not exist
○	1
○	$\frac{dx}{dy}$

Click to Save Answer & Move to Next Question

sec(s)

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 8 of 10 (Start time: 05:09:43 PM, 07 June 2021)

Total Marks: 1

Derivative of x is 1

Select the correct option

- [x] True
- [ ] False

MTH101:Quiz 1

Quiz Start Time: 05:06 PM, 07 June 2021

Question # 9 of 10 (Start time: 05:10:56 PM, 07 June 2021)

Total Marks: 1

$If\ f(x) = \tan^2(x^5)$, *then the derivative of* $f(x)$ *is* ______.

Select the correct option

Reload Math Equations

- ( ) None of these.
- (•) $10x^4 \sec^2(x^5)\tan(x^5)$.
- ( ) $-10x^4 \sec^2(x^5)\tan(x^5)$.
- ( ) $x^4 \sec^2(x^5)\tan(x^5)$.

MTH101:Quiz 1

Question # 10 of 10 (Start time: 05:11:25 PM, 07 June 2021)

pi radians corresponds to.........degrees.

Select the correct option

○	75
○	360
●	180
○	90

Click to

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 5 of 10 (Start time: 10:02:40 PM, 06 June 2021)

Total Marks: 1

$$What\ is\ the\ derivative\ of\ (3x^2+1)^{52}?$$

Select the correct option

Reload Math Equations

- ○ $(6x)^{52}$
- ● $312x(3x^2+1)^{51}$
- ○ $52(3x^2+1)^{51}$
- ○ *None of these*

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 6 of 10 (Start time: 10:03:08 PM, 06 June 2021)

Total Marks: 1

If a function f is continuous at a point then it ________________ be differentiable at that point.

Select the correct option

- ○ Is necessarily
- ● May

Time Left sec(s)

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 7 of 10 (Start time: 10:03:27 PM, 06 June 2021)

Total Marks: 1

If $f(x) = x \cos(x)$, then which of the following is true about it?

Select the correct option

- ● Its derivative with respect to x is $-x\sin(x) + \cos(x)$.
- ○ Its derivative with respect to x is $-x\sin(x) + \sin(x)$.
- ○ None of these.
- ○ Its derivative with respect to x is $x\sin(x) + \cos(x)$.

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 8 of 10 (Start time: 10:03:47 PM, 06 June 2021)

Total Marks: 1

If y=f(x) then instantaneous rate of change of y with respect to x at the point x0 is given by slope of tangent line at (x0,f(x0))

Select the correct option

- ○ False
- ● True

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 9 of 10 (Start time: 10:04:04 PM, 06 June 2021)

Total Marks: 1

x^0 where $x \neq 0$ is equal to $\ldots\ldots\ldots$

Select the correct option

Reload Math Equations

- [x] 1
- [ ] None of these
- [ ] 0
- [ ] ∞

MTH101:Quiz 1

Quiz Start Time: 10:01 PM, 06 June 2021

Question # 10 of 10 (Start time: 10:04:25 PM, 06 June 2021)

Total Marks: 1

If y = x^2, then which of the following is true about it over the interval [3, 4]
NOTE:- where x^n denotes the nth power of x.

Select the correct option

- Its average rate of change is 7.
- Its average rate of change is 9.
- None of these.
- Its average rate of change is 8.

Time Left sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 1 of 10 (Start time: 08:31:09 AM, 06 June 2021)

Total Marks: 1

The derivative of x^3 *is equal to..........?*

Select the correct option

Reload Math Equations

○	$3x^3$
●	$3x^2$
○	x
○	1

Question # 2 of 10 (Start time: 08:31:35 AM, 06 June 2021) Total Marks: 1

Derivative of absolute value function:
f(x) = Abs(x) = |x| at x = -2 is

Select the correct

- -1
- does not exist
- zero
- -2

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 3 of 10 (Start time: 08:32:06 AM, 06 June 2021)

Total Marks: 1

f(x) = Cosx is differentiable in the interval ______.

Select the correct option

- (-inf, 0]
- (-inf,+inf)
- [0,+inf)
- All are true

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 4 of 10 (Start time: 08:32:27 AM, 06 June 2021)

Total Marks: 1

$$What\ is\ the\ derivative\ \tan(x^2+x^3)?$$

Select the correct option

Reload Math Equations

- $(2x^3+3x)\sec^2(x^2+x^3)$
- $(2x^3+3x)\cos ec^2(x^2+x^3)$
- $(3x^2+2x)\sec^2(x^2+x^3)$
- None of these

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 5 of 10 (Start time: 08:33:08 AM, 06 June 2021)

Total Marks: 1

If y = x^2, then which of the following is true about it over the interval [3, 4]
NOTE:- where x^n denotes the nth power of x.

Select the correct option

- (●) Its average rate of change is 7.
- ( ) None of these.
- ( ) Its average rate of change is 8.
- ( ) Its average rate of change is 9.

Question # 6 of 10 (Start time: 08:33:33 AM, 06 June 2021)

Total Marks: 1

$$At\ x=\pi,\ the\ derivative\ of\ f(x)=a\cos x-b\sin x\ is$$
$$(note:\pi=180 degrees)$$

Select the correct option

Reload Math Equations

- -b
- a
- b
- -a

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 7 of 10 (Start time: 08:33:59 AM, 06 June 2021)

Total Marks: 1

$$If\ f(x) = (x^2 + \sin(x) + \cos(x))^{10},\ then\ the\ derivative\ of\ f(x)\ is\ \_\_\_\_\_.$$

Select the correct option

Reload Math Equations

- $10(x^2 + \sin(x) + \cos(x))^9$
- *None of these.*
- $10(x^2 + \sin(x) + \cos(x))^9(2x + \cos(x) - \sin(x))$
- $10(x^2 + \sin(x) + \cos(x))^9(2x + \cos(x) + \sin(x)).$

MTH101:Quiz 1

Quiz Start Time: 08:31 AM, 06 June 2021

Question # 8 of 10 (Start time: 08:34:32 AM, 06 June 2021)

Total Marks: 1

Which of the following rule will be used to find the derivative of $x\cot x$?

Select the correct option

Reload Math Equations

- sum
- power
- product
- difference

sec(s)

MTH101:Quiz 1 Quiz Start Time: 08:31 AM, 06 June 2021

Question # 9 of 10 (Start time: 08:35:02 AM, 06 June 2021) Total Marks: 1

If (0,0) and (2,2) are any two pints on a curve then the average speed of a particle passing through this curve from (0,0) to (2,2) is --------

Select the correct option

- -1
- 2
- -2
- 1

MTH101:Quiz 1

Question # 10 of 10 (Start time: 08:35:24 AM, 06 June 2021)

Simplifed expression for f(x)=tanx/secx is

Select the correct option

- Sinx
- Cosx

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 1 of 10 (Start time: 08:20:35 AM, 06 June 2021)

Total Marks: 1

For the curve: f(t) = -2t+3, the instantaneous rate of change of 'f(t)' at 't = a' is ______.

Select the correct option

- -1
- -2
- 1
- 2

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 2 of 10 (Start time: 08:21:05 AM, 06 June 2021)

Total Marks: 1

Slope of secant line joining points (1.1) and (3.4) is____.

Select the correct option

- 1.5
- 1
- 1.3
- None of these

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 3 of 10 (Start time: 08:22:27 AM, 06 June 2021)

Total Marks: 1

$$If\ f(x) = secx\ and\ g(x) = csc(x),\ then\ the\ derivative\ of\ f(x) \times g(x)\ is$$

Select the correct option

Reload Math Equations

- $-\sec x \csc x \cot x - \csc x \sec x \tan x$
- None of these.
- $-\sec x \csc x \cot x + \csc x \sec x \tan x$
- $\sec x \csc x \cot x + \csc x \sec x \tan x$

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 4 of 10 (Start time: 08:23:18 AM, 06 June 2021)

Total Marks: 1

Derivative of a straight line parallel to x-axis is------

Select the correct option

- -1
- Infinity
- +1
- Zero

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 5 of 10 (Start time: 08:23:57 AM, 06 June 2021)

Total Marks: 1

$(f.g)' = f'.g - f.g'$

Select the correct option

True

False

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 6 of 10 (Start time: 08:24:26 AM, 06 June 2021)

Total Marks: 1

The graph of function y=2 is a.....line with slope 0,is

Select the correct option

- [x] Horizontal
- [ ] Vertical

Saving

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 7 of 10 (Start time: 08:25:16 AM, 06 June 2021)

Total Marks: 1

$$\text{If } u(t) = 4\cos t\,,\ then\ u'(\frac{\pi}{2}) = \ldots$$

Select the correct option

Reload Math Equations

○	2
○	9
●	-4
○	4

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 8 of 10 (Start time: 08:26:00 AM, 06 June 2021)

Total Marks: 1

$$If\ f(x) = x^5\ and\ g(x) = x^2,\ then\ the\ derivative\ of\ \frac{f(x)}{g(x)}\ is$$

Select the correct option

Reload Math Equations

- 3x
- $3x^2$
- None of these
- $\frac{3}{x^2}$

MTH101:Quiz 1

Quiz Start Time: 08:20 AM, 06 June 2021

Question # 9 of 10 (Start time: 08:26:26 AM, 06 June 2021)

Total Marks: 1

$What\ is\ the\ derivative\ of\ \cot(x^6)?$

Select the correct option

Reload Math Equations

○	$\cos ec(x^6)\cot(x^6)$
○	None of these
●	$-6x^5 \cos ec^2(x^6)$
○	$-\cos ec(x^6)$

MTH101:Quiz 1

Question # 10 of 10 (Start time: 08:26:57 AM, 06 June 2021)

$If\ f(x) = x^5\ and\ g(x) = x^2,\ then\ the derivative\ of\ f(x) \times g(x)\ is$

Select the correct option

- ○ $6x^3$
- ○ $3x^2$
- ● None of these
- ○ $7x^2$

MTH101:Quiz 1 Quiz Start Time: 08:10 AM, 06 June 2021

Question # 1 of 10 (Start time: 08:10:14 AM, 06 June 2021) Total Marks: 1

What is the derivative of $\cos(2x)$ at $x = \pi/4$?

Select the correct option

Reload Math Equations

○	0
○	None of these
●	-2
○	2

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 2 of 10 (Start time: 08:11:02 AM, 06 June 2021)

Total Marks: 1

d(-2x+3cosx)/dx=..

Select the correct option

2-3sinx

-2-3sinx

-2+3sinx

2+3sinx

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 3 of 10 (Start time: 08:11:43 AM, 06 June 2021)

Total Marks: 1

If y = x^2, then which of the following is true about it over the interval [3, 4]
NOTE:- where x^n denotes the nth power of x.

Select the correct option

- None of these.
- Its average rate of change is 9.
- Its average rate of change is 7.
- Its average rate of change is 8.

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 4 of 10 (Start time: 08:12:14 AM, 06 June 2021)

Total Marks: 1

If in the graph of f(x) , there comes a corner then the function is -------- at that corner.

Select the correct option

- ○ Differentiable
- ● Non differentiable

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 5 of 10 (Start time: 08:12:46 AM, 06 June 2021)

Total Marks: 1

$$If\ y = f(\sin t)\ and\ y = t\ then\ \left(\frac{df}{dy}\right) = \ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

Select the correct option

Reload Math Equations

- $\sec y$
- $-\sec y$
- $\cos y$
- $-\cos y$

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 6 of 10 (Start time: 08:13:17 AM, 06 June 2021)

Total Marks: 1

Which of the followings is the derivative of f(x) = 2log x?

Select the correct option

Reload Math Equations

- $f'(x) = \frac{2}{x^2}$
- $f'(x) = \frac{1}{x ln 2}$
- $f'(x) = \frac{1}{\sqrt{x}}$
- $f'(x) = \frac{2}{x}$

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 7 of 10 (Start time: 08:14:01 AM, 06 June 2021)

Total Marks: 1

A bus travels 175 miles in 2.5 hours. Its average velocity will be.......

Select the correct option

- 80m/h
- 90m/h
- 70m/h
- 60m/h

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 8 of 10 (Start time: 08:14:38 AM, 06 June 2021)

Total Marks: 1

f $f(x) = \tan x^2$ *and* $g(x) = \sin x^{10}$, *then the derivative of* $f(x) + g(x)$ *is* . . . .

Select the correct option

Reload Math Equations

- [ ] $\sec^2(x^2) + \cos x^{10}$
- [ ] $2x\sec(x^2) + 10x^9\cos x^{10}$
- [x] $2x\sec^2(x^2) + 10x^9\cos x^{10}$
- [ ] None of these

sec(s)

MTH101:Quiz 1

Quiz Start Time: 08:10 AM, 06 June 2021

Question # 9 of 10 (Start time: 08:15:36 AM, 06 June 2021)

Total Marks: 1

$Derivative\ of\ \sin(5x+3)\ is\ \ldots\ldots\ldots\ldots\ldots\ldots$

Select the correct option

Reload Math Equations

○	$3\cos(5x+3)$
○	$3\cos 5x$
○	$-5\sin(5x+3)$
	$5cox(5x+3)$

MTH101:Quiz 1

Question # 10 of 10 (Start time: 08:16:13 AM, 06 June 2021)

Secx is equal to the reciprocal of

Select the correct option

- Cosecx
- Tanx
- Cosx
- Sinx